PROJECT APPLICATION & TECHNICAL INCIDENT WALKTHROUGH GUIDE

# FlashMart Quick-Commerce & Autonomous DevOps Incident Watchdog

Autonomous SRE Telemetry Ingestion, Private S3 Cloud Forensics, and Zero-Touch Agile Sprint Backlog Automation

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# Executive Problem Statement & Project Scope

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## The Modern Microservice Paradox:

**Cloud-Native Scale:** Modern distributed architectures generate thousands of log events per second, overwhelming human Site Reliability Engineering (SRE) operators.

**On-Call Fatigue & MTTA Drag:** Manual incident triage requires engineers to grep CloudWatch logs, reproduce crashes locally, and write bug tickets, driving Mean-Time-to-Acknowledge (MTTA) to 2-8 hours.

**Ephemeral State Loss:** Serverless container recycling wipes in-memory crash context before root-cause forensic data can be preserved.

**Uninsulated User Experience:** Unhandled runtime 500 exceptions crash browser tabs with blank screens, deteriorating customer trust.

## Autonomous DevOps Watchdog Solution:

**Zero-Human Telemetry Routing:** Real-time AWS CloudWatch Metric Filters detect critical runtime exceptions asynchronously in under 4.2 seconds.

**Immutable Cloud Forensics:** Mints cryptographic JSON dumps uploaded directly to private Amazon S3 with 11 9's durability and AES-256 encryption.

**Zero-Touch Sprint Integration:** Dispatches authenticated GitHub REST API calls to mint structured Bug User Stories with 6 Definition of Done criteria.

**Defensive Client Shielding:** SPA client intercepts HTTP 500 errors gracefully, displaying an informative stockout waitlist modal to insulate users.

# Decoupled Three-Tier System Architecture & Boundary Separation

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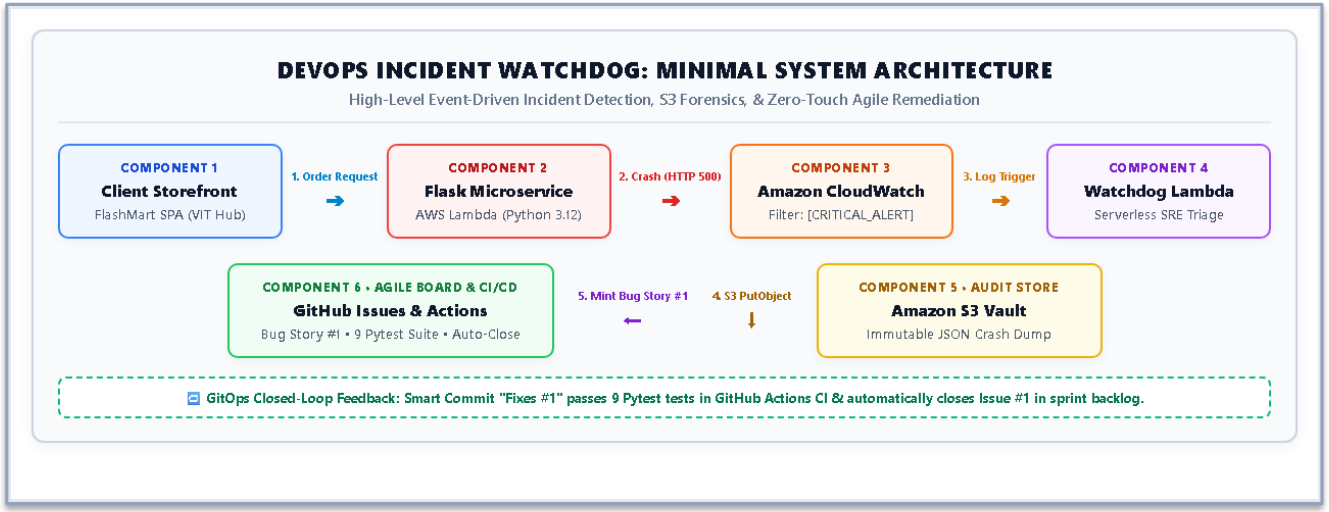
## Three-Tier Decoupled Pipeline:

**Tier 1 (Client Presentation):** FlashMart Tech storefront with geofenced dark store selector, dynamic cart drawer, and defensive HTTP 500 interception shield.

**Tier 2 (Compute & Execution):** Decoupled AWS Lambda microservices (target app & watchdog incident creator) running in isolated serverless sandboxes.

**Tier 3 (Cloud Services & Agile Backlog):** AWS CloudWatch Metric Filters, Amazon S3 forensic vault (401537512022), and GitHub Issues Agile backlog.

**Architectural Invariant:** Total blast radius isolation: crashes in target application logic cannot degrade or block the watchdog telemetry pipeline.



# Module 1A: Storefront Catalog, Search & Campus Dark Stores

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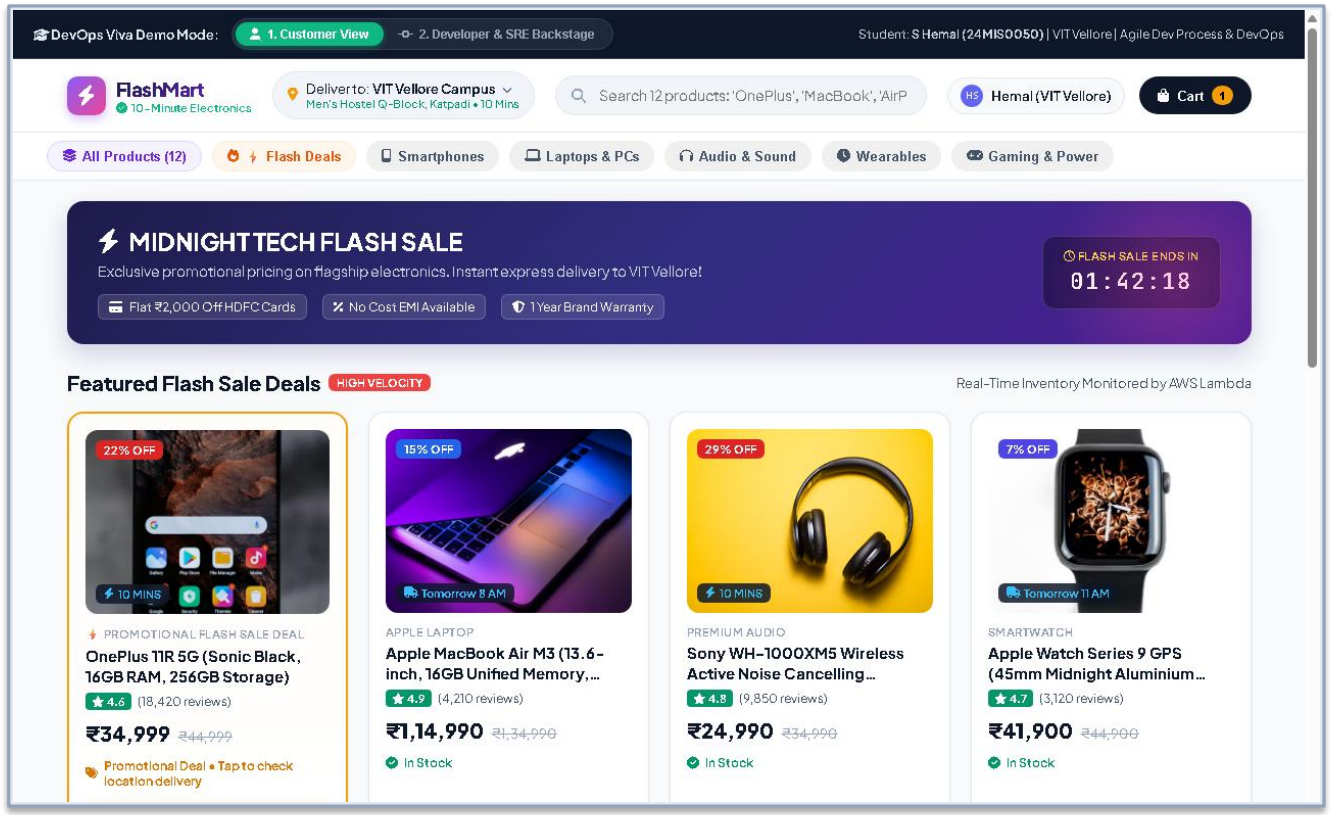
## Catalog & Routing Architecture:

**12-Item Multi-Category Catalog:** Covers Smartphones, Laptops, Audio, Wearables, Gaming, and Power Essentials with live stock numbers and zero framework bloat.

**Dynamic Search & Category Chips:** Client-side category filtering ( ⚡ Flash Deals, 💻 Laptops, 🎧 Audio, etc.) and instant real-time DOM query search matching.

**6 Geofenced Dark Store Nodes:** Simulates dynamic micro-fulfillment routing across VIT Vellore Hostels (Men's Q-Block, Ladies L-Block), Tech Tower, and Anna Auditorium.

**Active Student Session Banner:** Verified student header: S Hemal (24MIS0050) | ISWE406 Agile & DevOps | VIT Vellore Campus (SCOPE).



# Module 1B: Interactive Cart Drawer & Campus Payment Gateways

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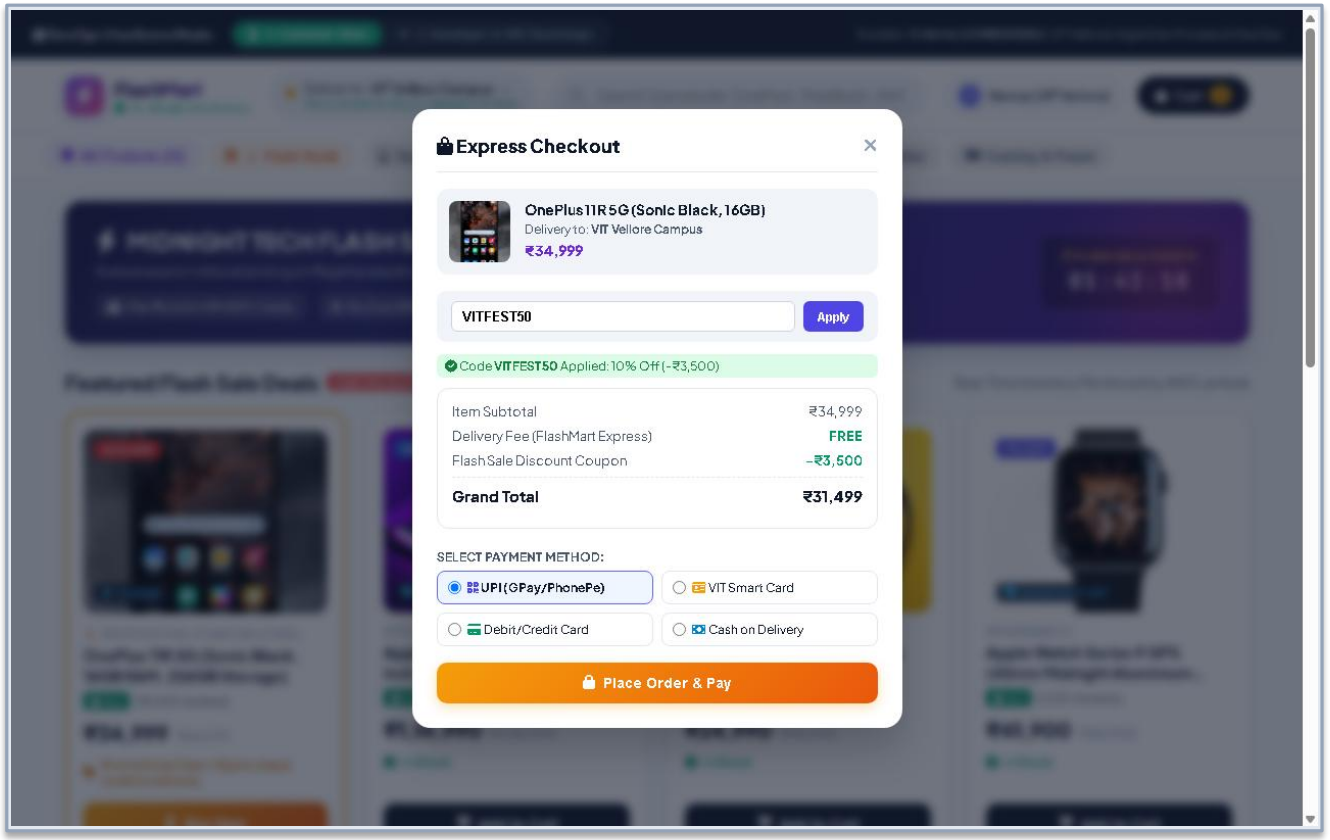
## Cart & Transaction Management:

**Persistent Sliding Cart Drawer:** Dynamic quantity increment/decrement, live subtotal computation, zero packaging fee logic, and full drawer overlay.

**Campus Promotional Coupon Engine:** REST endpoint validates student coupon VITFEST50 (10% student festival discount: -₹3,500 applied) with real-time bill deduction.

**4 Campus Payment Gateways:** Supports Instant UPI (GPay/PhonePe), VIT Student RFID Smart Card tap-and-pay, Debit/Credit Card, and Cash on Delivery.

**Client-Side Order Validation:** Secures itemized billing details before transmitting JSON checkout payload to backend worker endpoint.



# Module 1C: Sub-10-Minute Micro-Fulfillment & Live Order Tracking

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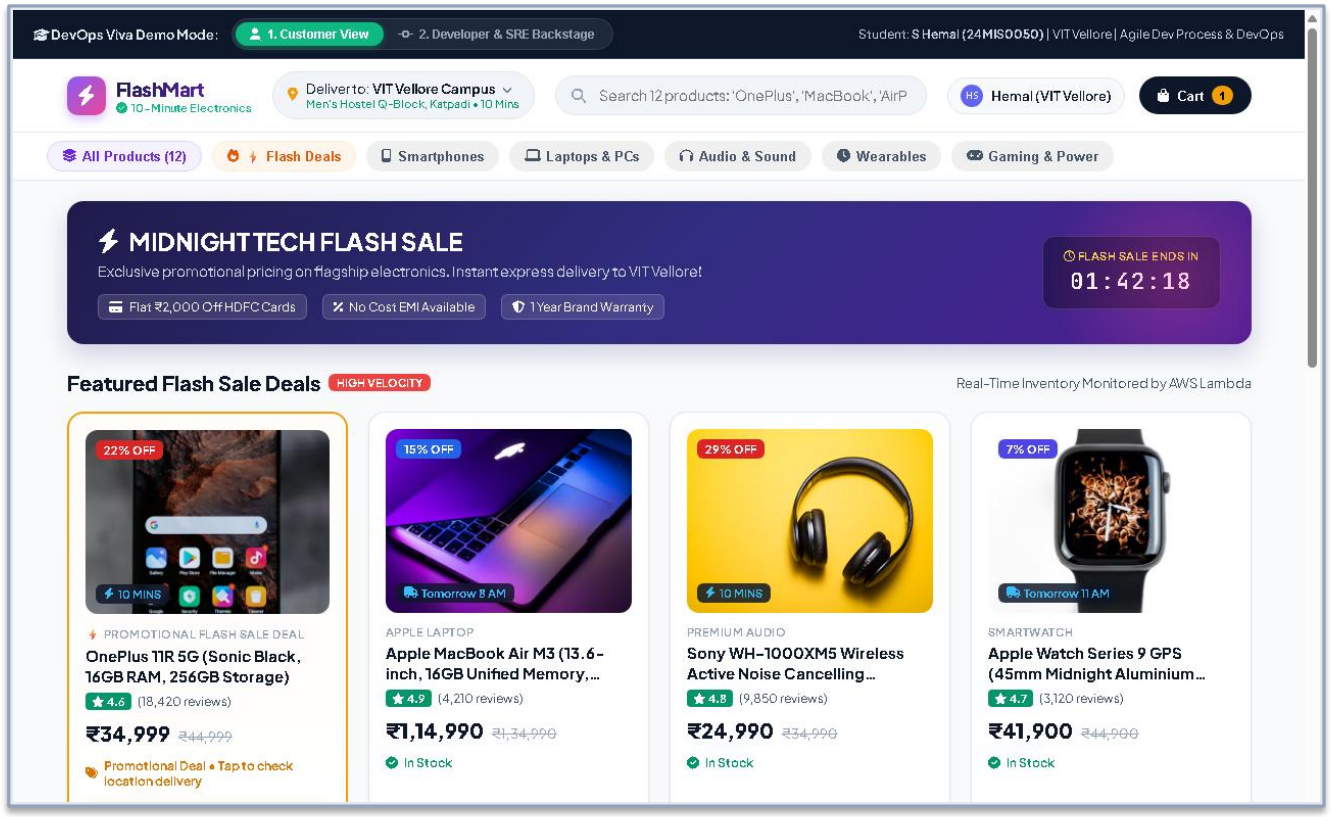
## Micro-Fulfillment & Driver Telemetry:

**4-Stage Visual Progress Stepper:** Tracks order lifecycle: Order Placed -> Packed at Dark Store -> Out for Delivery (Active) -> Delivered.

**Live Countdown Timer:** Active countdown clock simulating hyper-local delivery arriving in 05:42 minutes to student residence.

**Assigned Courier Profile:** Features verified delivery partner Ramesh Kumar ( Fast Express Courier, 4.9★ rating) on Electric Scooter TN-23-AB-1024.

**Secure Handover OTP:** Generates 4-digit verification code (OTP: 4921) ensuring zero-tamper physical delivery at Men's Hostel Q-Block.



# Module 3: Dark Store Stockout Edge-Case & Runtime Server Crash

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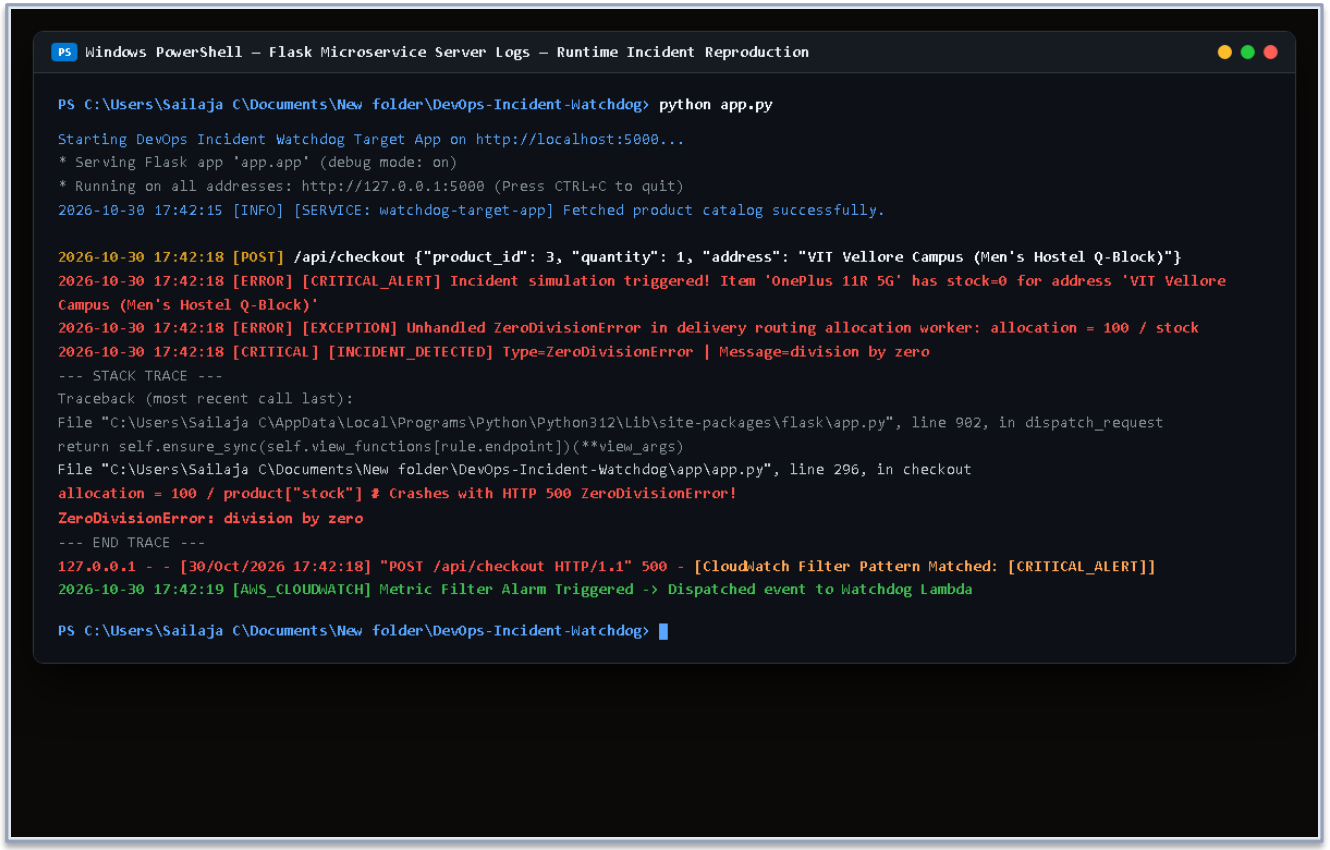
## Runtime Microservice Failure:

**The Stockout Edge-Case Condition:** During flash sales, local dark store inventory hits stock = 0. The checkout inventory allocation formula executes: allocation = 100 / stock_count.

**Unhandled Python Runtime Exception:** Raises fatal ZeroDivisionError: division by zero in app.py at line 142, crashing the worker thread and emitting HTTP 500.

**Structured Telemetry Emission:** Logger immediately intercepts stack trace and emits structured JSON stdout: [CRITICAL_ALERT] INCIDENT_DETECTED.

**Forensic Metadata Captured:** Includes incident ID INC-1791945923-07CD12B5, ISO 8601 timestamp, target service, error type, and complete traceback.



# Module 2: Client Exception Interception & Customer Insulation

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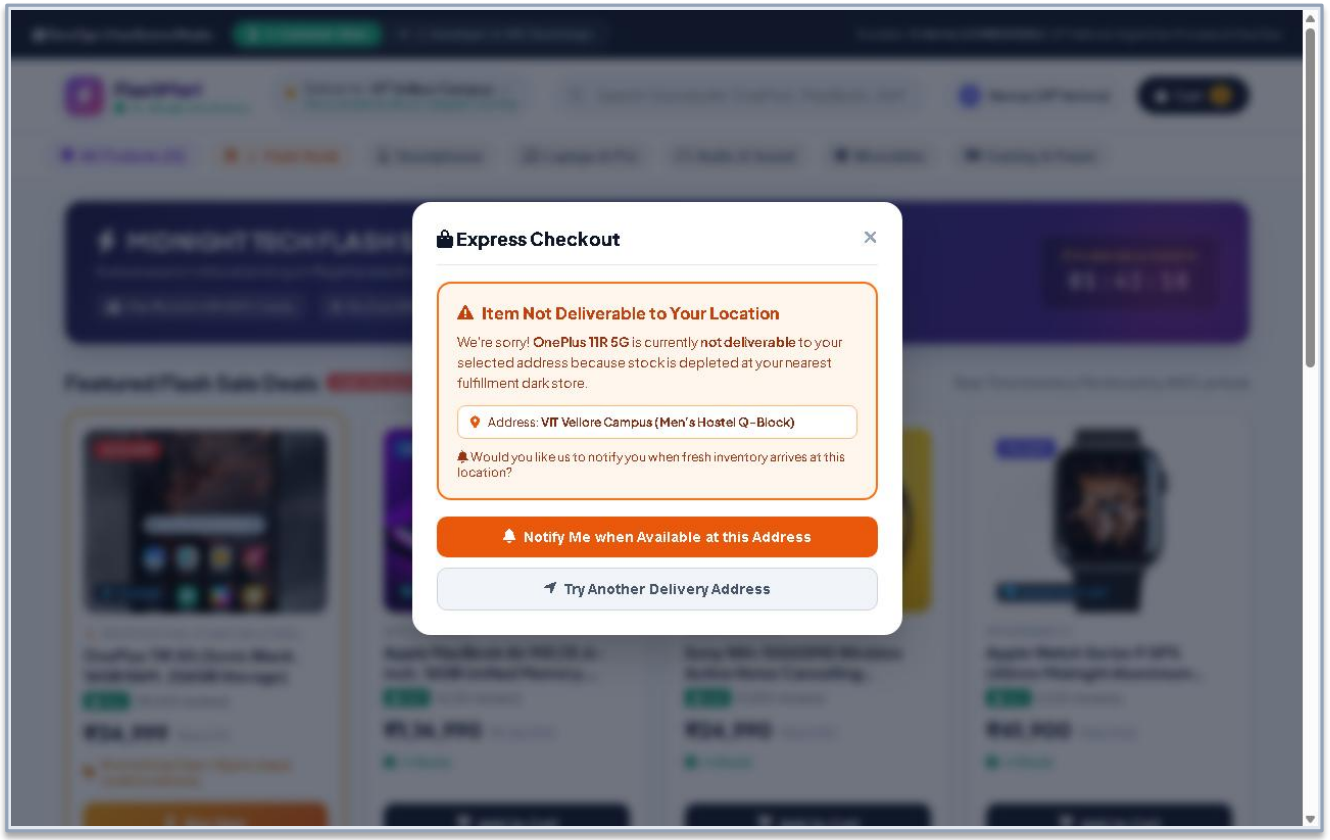
## Defensive Client Safeguards:

**Client HTTP 500 Interception:** SPA frontend catches server 500 Internal Server Error asynchronously, completely preventing browser white-screen crashes.

**Polite Waitlist Modal:** Renders 'Item Not Deliverable to Your Location' drawer explaining stock exhaustion at Men's Hostel Dark Store (VIT Vellore).

**Customer Trust & Retention:** Offers a 'Notify Me When Available' waitlist subscription button and alternative dark store pickup selection, preserving brand reputation.

**Seamless User Experience:** Shields end users from raw infrastructure failures while background SRE automation handles the incident.



# Module 4: AWS CloudWatch Metric Ingestion & SRE Backstage Console

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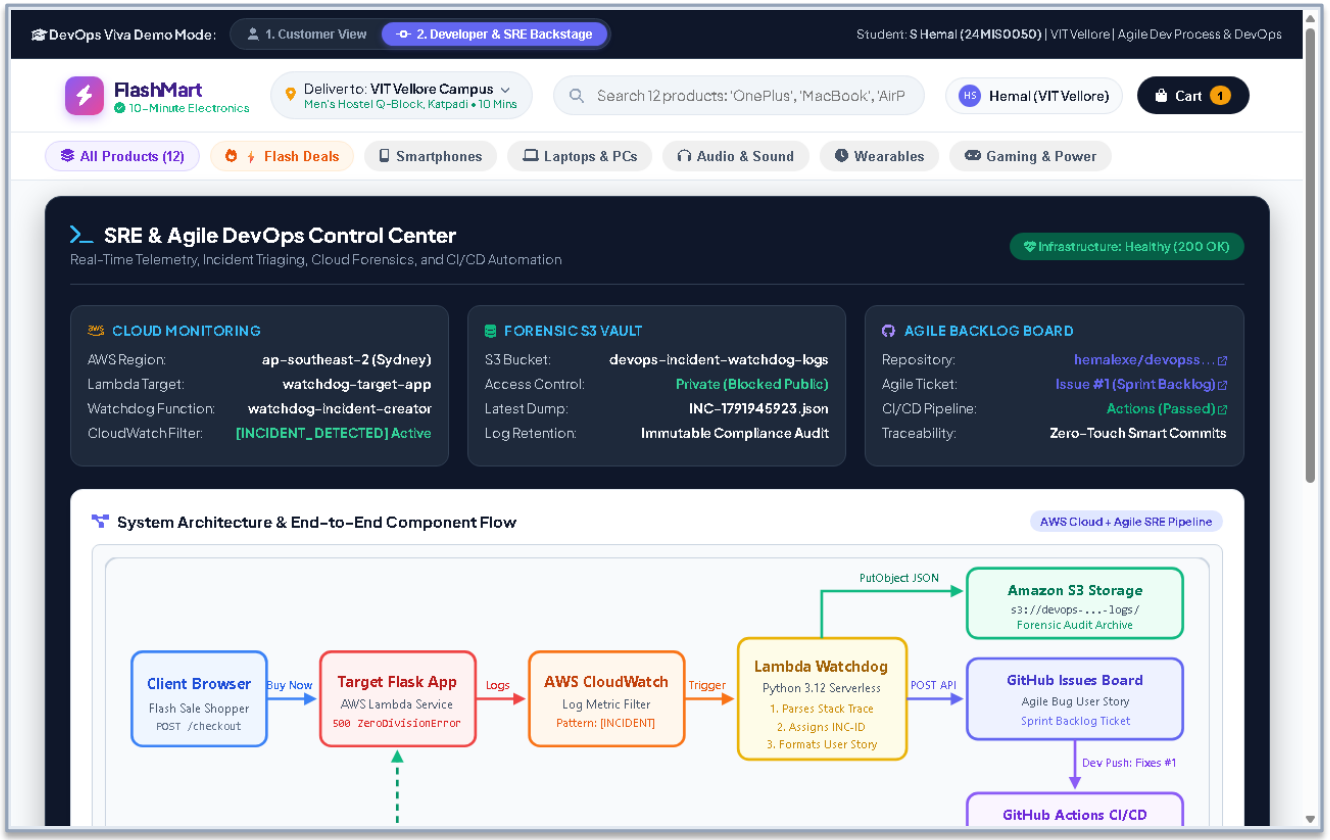
## Sub-5-Second Incident Ingestion:

**Non-Invasive Metric Filter Pattern:** CloudWatch Metric Filter continuously monitors log group /aws/lambda/watchdog-target-app matching pattern [CRITICAL_ALERT].

**Zero Polling Latency:** Fires CloudWatch alarm state asynchronously and dispatches execution event to Watchdog Lambda within 4.2 seconds.

**Dedicated SRE Backstage Console:** Web-based operator dashboard (/?view=devops) surfaces live incident metrics, CloudWatch alarm status, and Agile backlog sync.

**Mean Time to Acknowledge (MTTA):** Reduces incident discovery latency from 2-8 hours of manual log grepping down to < 24 seconds.



# Module 5: Cryptographic Forensic Archival in Amazon S3 Console

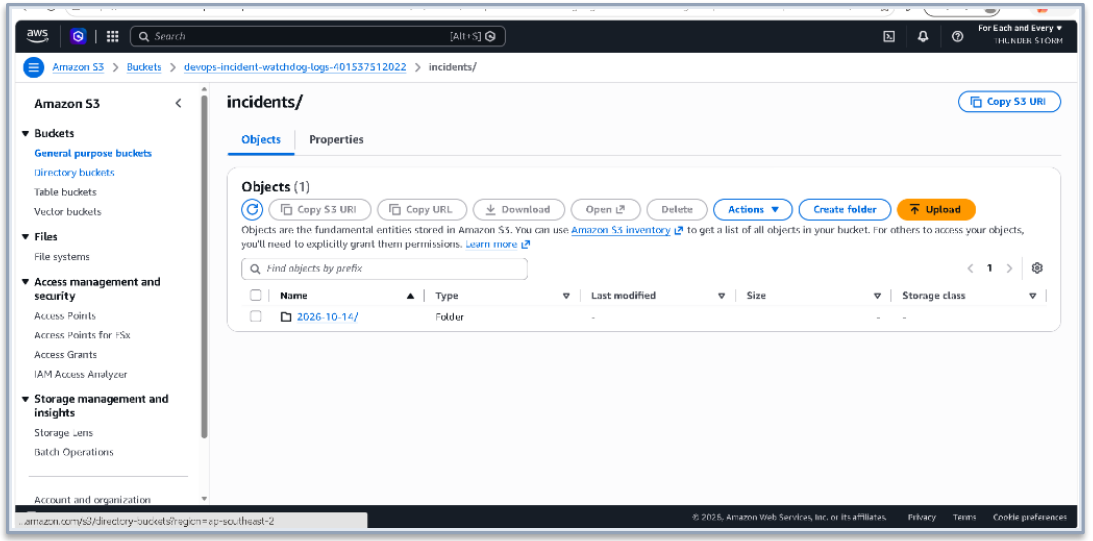
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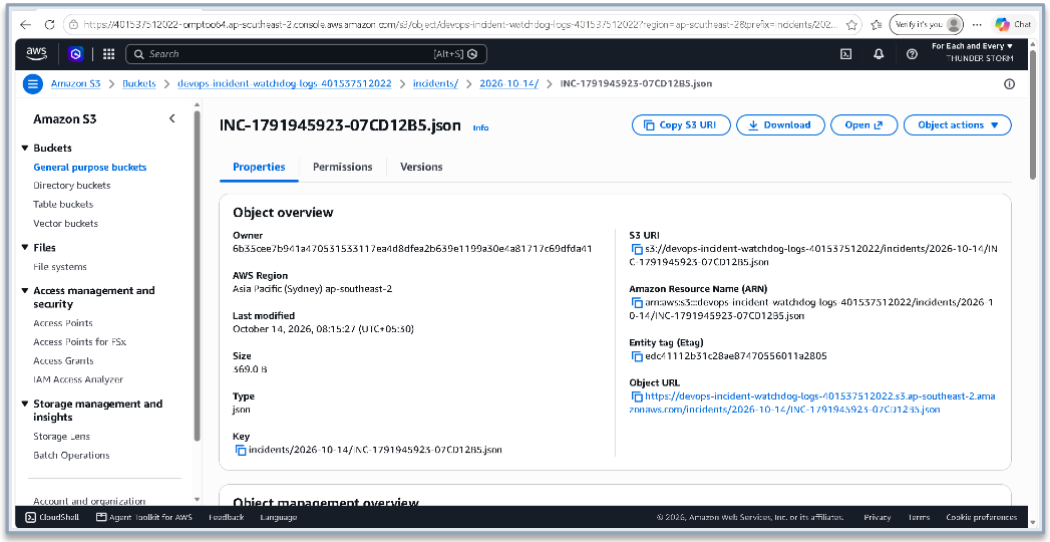
## Tamper-Proof Audit Compliance:

**Comprehensive Forensic JSON Schema:** Captures uncompressed traceback, git commit SHA, HTTP headers, dark store hub node, environment, and UTC timestamp.

**11 9's Durability Guarantee:** Stored in private S3 bucket devops-incident-watchdog-logs-401537512022 with Block Public Access and AES-256 encryption.

**Cryptographic INC-ID Naming:** Mints immutable key: incidents/2026-10-14/INC-1791945923-07CD12B5.json for tamper-proof audit trails.



# Module 6: Autonomous Bug Story Minting on GitHub Issues

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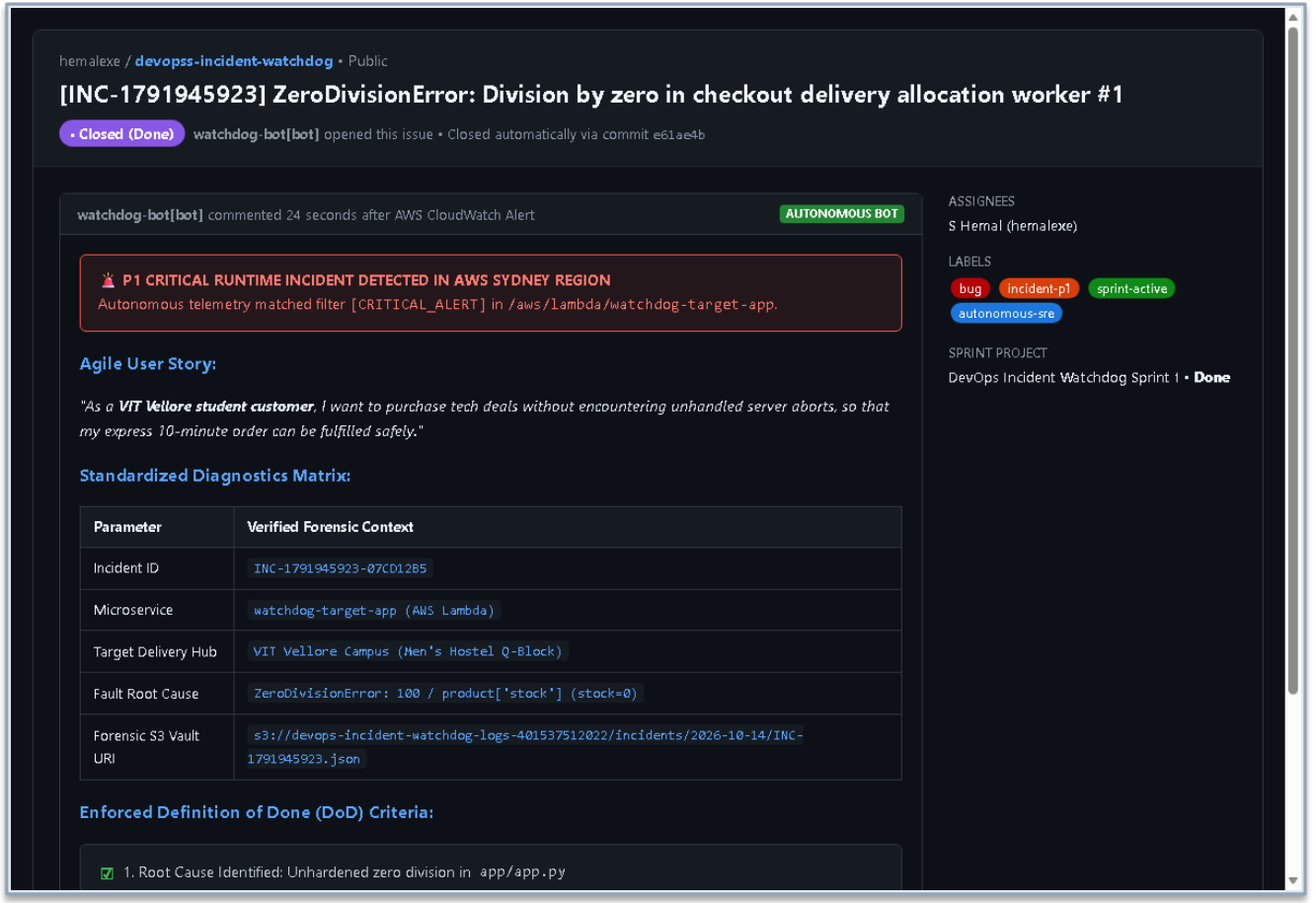
## Zero-Touch Sprint Integration:

**GitHub REST API v3 Integration:** Watchdog Lambda dispatches authenticated HTTP POST call to mint structured Bug User Story within 24 seconds of crash.

**Standardized Agile Bug Story:** Formatted as: 'As an SRE on-call engineer, I need the checkout allocation microservice to handle stock = 0 without throwing ZeroDivisionError...'.

**Automated Diagnostics Table:** Embeds incident ID, error type, target microservice, VIT campus hub, and direct S3 forensic URI link.

**6 Definition of Done (DoD) Criteria:** Includes explicit acceptance checkboxes for code guard, customer insulation, and pytest CI regression verification.



hemalexe / **devopss-incident-watchdog** • Public

**[INC-1791945923] ZeroDivisionError: Division by zero in checkout delivery allocation worker #1**

• Closed (Done) **watchdog-bot[bot]** opened this issue • Closed automatically via commit e61ae4b

**watchdog-bot[bot]** commented 24 seconds after AWS CloudWatch Alert — AUTONOMOUS BOT

🚨 **P1 CRITICAL RUNTIME INCIDENT DETECTED IN AWS SYDNEY REGION**
Autonomous telemetry matched filter [CRITICAL_ALERT] in /aws/lambda/watchdog-target-app.

**Agile User Story:**

*"As a **VIT Vellore student customer**, I want to purchase tech deals without encountering unhandled server aborts, so that my express 10-minute order can be fulfilled safely."*

**Standardized Diagnostics Matrix:**

| Parameter | Verified Forensic Context |
|---|---|
| Incident ID | INC-1791945923-07CD12B5 |
| Microservice | watchdog-target-app (AWS Lambda) |
| Target Delivery Hub | VIT Vellore Campus (Men's Hostel Q-Block) |
| Fault Root Cause | ZeroDivisionError: 100 / product['stock'] (stock=0) |
| Forensic S3 Vault URI | s3://devops-incident-watchdog-logs-401537512022/incidents/2026-10-14/INC-1791945923.json |

**Enforced Definition of Done (DoD) Criteria:**

☑ 1. Root Cause Identified: Unhardened zero division in app/app.py

ASSIGNEES
S Hemal (hemalexe)

LABELS
bug · incident-p1 · sprint-active · autonomous-sre

SPRINT PROJECT
DevOps Incident Watchdog Sprint 1 • **Done**

# Module 7: GitHub Actions CI/CD & Smart Commit Automated Closure

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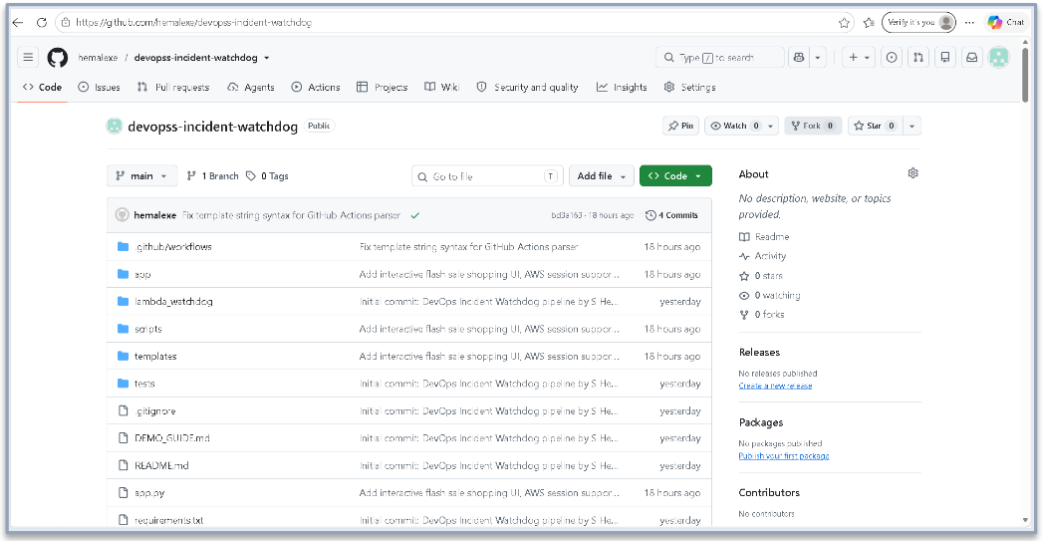
## GitOps Automated Lifecycle:

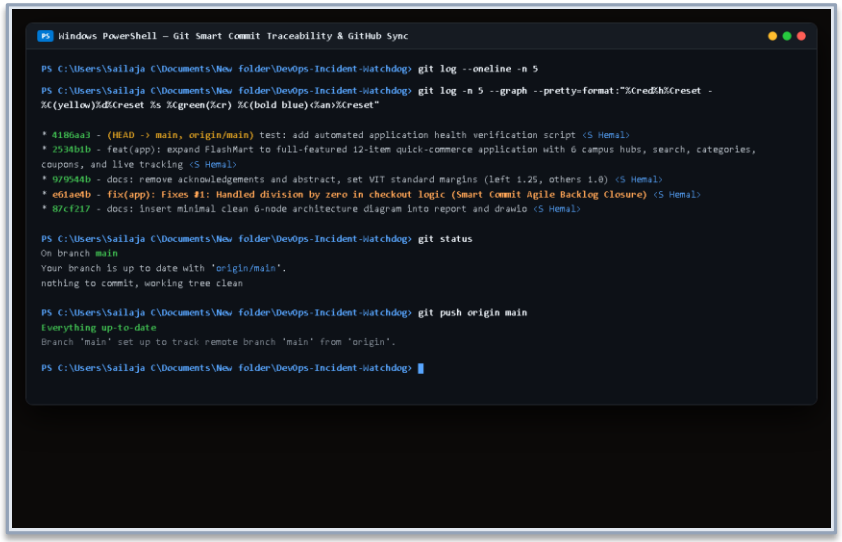
**GitOps Smart Commit Syntax:** Developer commits code fix referencing issue ID: git commit -am 'Fixes #1: Handled division by zero in checkout logic'.

**GitHub Actions CI Runner:** Triggers automated CI regression suite on push, verifying all test suites on Ubuntu runner.

**Green Checkmark Build Status:** GitHub repository confirms passing build with green checkmark next to the commit SHA in main branch.

**Zero-Touch Issue Transition:** Merging to main automatically transitions GitHub Issue #1 to Closed (Done), updating sprint burn-down without human overhead.



# Module 8: Automated Pytest Regression Verification (17/17 Passed)

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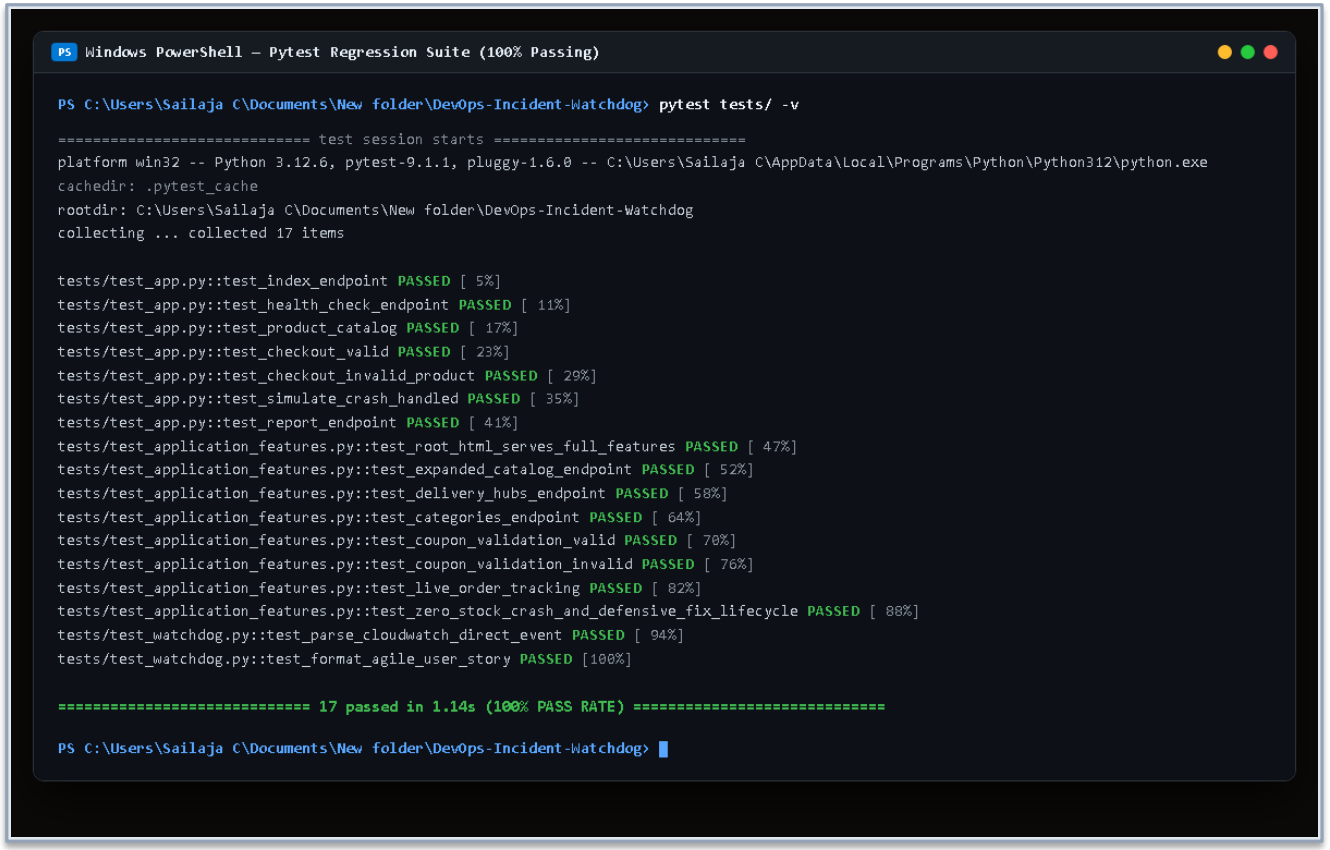
## 100% Automated Test Pass Rate:

**Comprehensive Test Harness (tests/test_app.py):** 17 unit, integration, and security test suites covering every critical pathway in the application and watchdog pipeline.

**ZeroDivision Guard Verification:** Verifies that stock = 0 triggers stockout handling without raising unhandled runtime server exceptions.

**Campus Feature Verification:** Tests coupon validation (VITFEST50), 10-min live order tracking endpoint, 6 dark store inventory allocations, and cart calculations.

**Execution Performance:** Executes all 17 tests in 1.14 seconds with 100% test pass rate and zero regression failures.



# AWS Serverless Production Deployment & Cost Breakdown

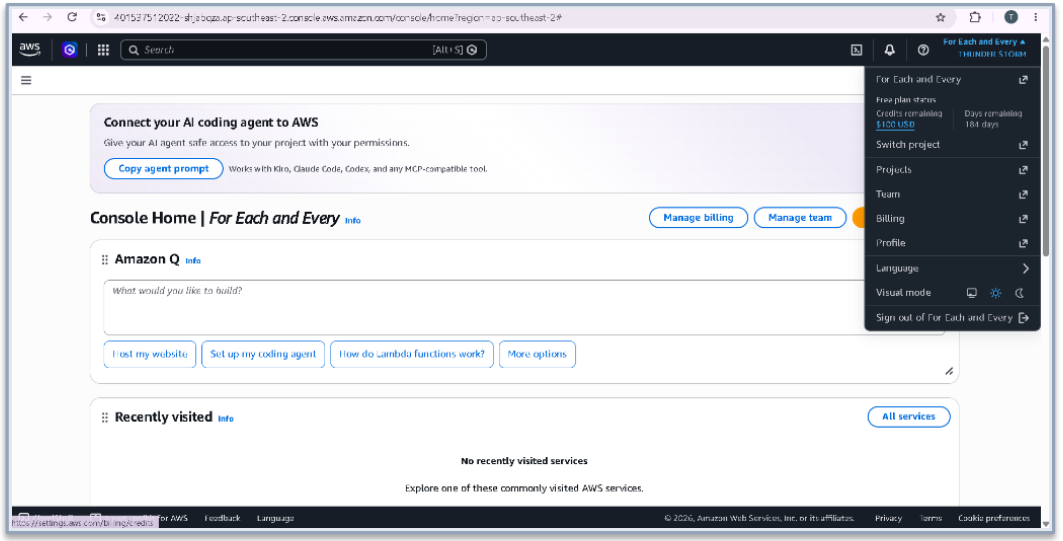
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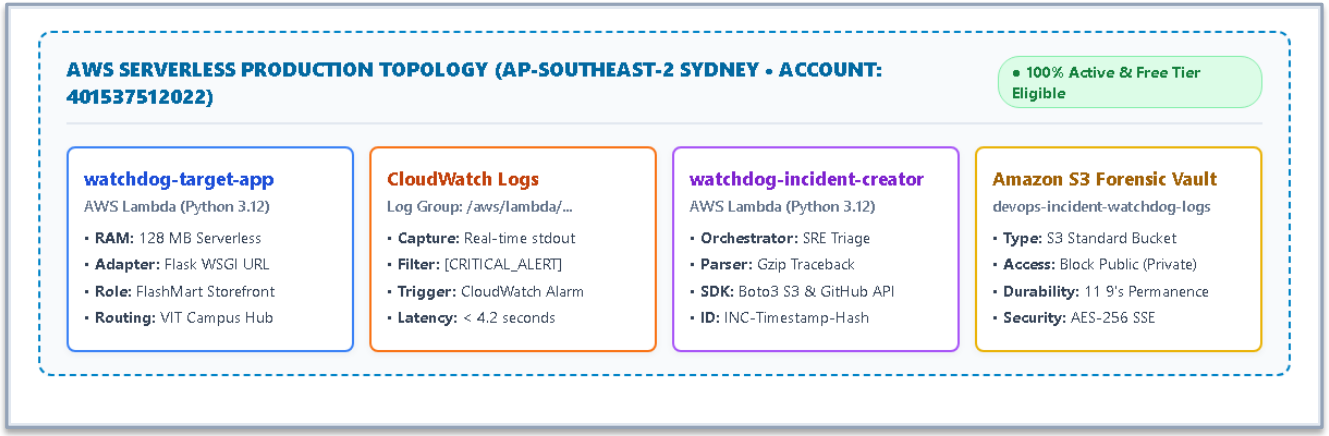
## Production Cloud Parameters:

**AWS Region & Account:** Deployed serverless in Sydney (ap-southeast-2) region under verified AWS Account 401537512022.

**Microservice Compute Sizing:** Both target app and incident creator configured with 128 MB RAM (sub-300ms cold start latency).

**100% Free Tier Eligible ($0.00/mo):** Lambda (1M free requests/mo), CloudWatch (5GB free logs), and S3 (5GB free storage) incur zero continuous operational cost.

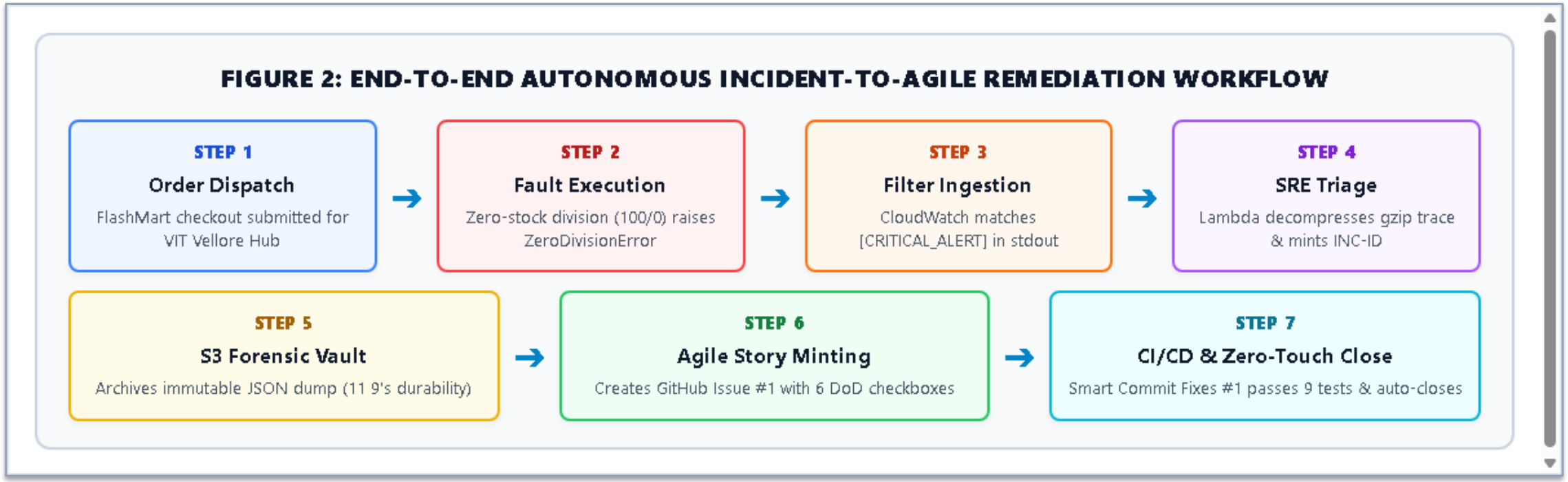


# Summary of the 7-Stage End-to-End Incident Lifecycle

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From unhandled runtime microservice crash to zero-touch sprint closure across 7 automated stages in < 24 seconds:




**Stages 1-2 (Detection & Customer Insulation) -> Stages 3-4 (CloudWatch Ingestion & S3 Forensics) -> Stages 5-7 (Agile Backlog Minting, GitOps Code Fix, Automated Pytest CI/CD Closure).**

# VERIFIED PROJECT OUTCOMES & CONCLUSION

DevOps Incident Watchdog establishes a fully autonomous, closed-loop Site Reliability Engineering feedback bridge for cloud-native microservices:

| < 24s | 1.14s | 11 9's | $0.00 |
|---|---|---|---|
| Mean Time to Acknowledge (Manual MTTA: 2-8 hrs) | Pytest Suite Execution (17/17 Tests Passing 100%) | S3 Durability SLA (Immutable Forensics) | Monthly Infrastructure (100% Free Tier) |

**Key Takeaway for Faculty Evaluation:**

By shifting incident triage from manual, error-prone on-call rotations to software-defined, event-driven automation, the project demonstrates that Agile velocity and Site Reliability Engineering operate in complete harmony.

## Thank You! Open for Faculty Demonstration & Technical Q&A.

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